



# 4JH Series - 80 / 110

|                                     |                                                         |                       |
|-------------------------------------|---------------------------------------------------------|-----------------------|
| Configuration                       | In-line, 4-stroke, vertical, water-cooled diesel engine |                       |
| Max. output at crankshaft*          | 58.8 kW@3200 rpm                                        | [80 mhp@3200 rpm]     |
|                                     | 80.9 kW@3200 rpm                                        | [110 mhp@3200 rpm]    |
| No. of cylinders                    | 4 cylinder                                              |                       |
| Displacement                        | 2.000 L                                                 | [1.22 cu in]          |
| Bore x stroke                       | 84 mm x 90 mm                                           | [3.31 in x 3.54 in]   |
| Aspiration                          | Turbocharged & intercooled                              |                       |
| Combustion system                   | Direct injection, Bosch common rail system              |                       |
| Starting system                     | Electric starting 12V - 1.4 kW                          |                       |
| Engine management                   | Electronic                                              |                       |
| Cooling system                      | Fresh water cooling by centrifugal fresh water pump     |                       |
|                                     | Seawater cooling by rubber impeller seawater pump       |                       |
| Alternator                          | 12V - 125A                                              |                       |
| Lubrication system                  | Enclosed, forced lubricating system                     |                       |
| Engine lube oil specifications      | API class: CD or higher, SAE grade: 10W-30 or 15W-40    |                       |
| Direction of rotation at crankshaft | Counterclockwise (CCW) viewed from stern                |                       |
| Dry weight without gear             | 229 kg                                                  | [505 lbs]             |
| Engine dimensions† L x W x H        | 909 x 613 x 675 mm                                      |                       |
| Emission compliance                 | EU: RCD 2, BSO II, EMC                                  | US: EPA Tier 3, EPA-C |
| Flywheel housing size               | SAE #4                                                  |                       |
| Controls                            | Mechanical (optional - VC20 system)                     |                       |

NOTE: Fuel condition: Density at 15°C = 0.84 g/cm<sup>3</sup>: 1 kW = 1.3596 mhp = 1.3410 HP

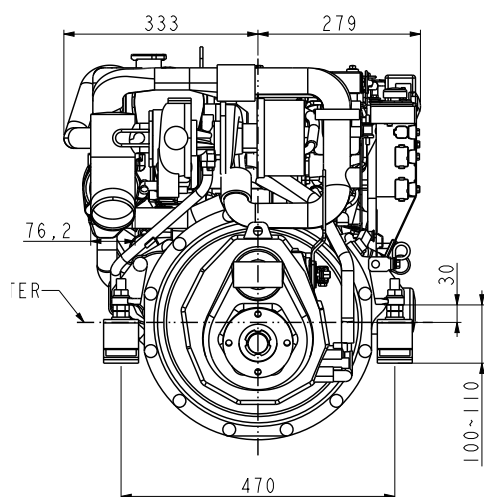
Technical data is according to (ISO 8665) / (ISO 3046)

\* Fuel temperature 40°C at the inlet of the fuel injection pump [ISO 8665: 2006]

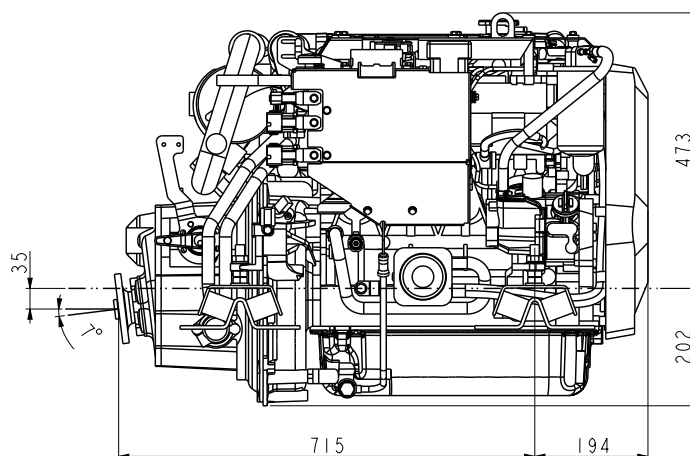
†Length includes KM4A2 marine gear (Dimensions may vary with other gears, drives or elbow)

## DIMENSIONS

**Rear view**



**Right side view**

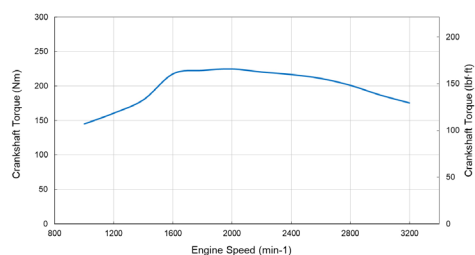


4JH80/110 with KM4A2 marine gear  
(Not to be used for engine installation)

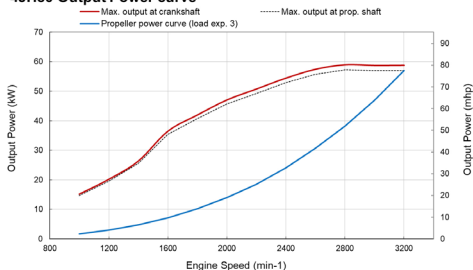


## PERFORMANCE CURVES

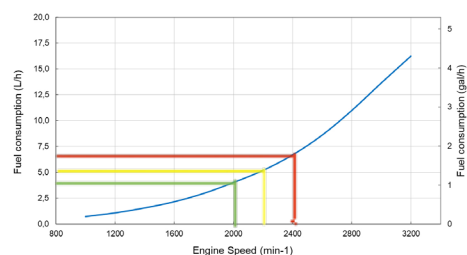
4JH80 Torque curve



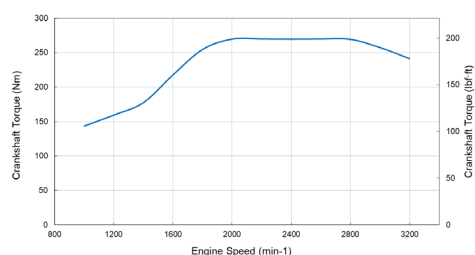
4JH80 Output Power curve



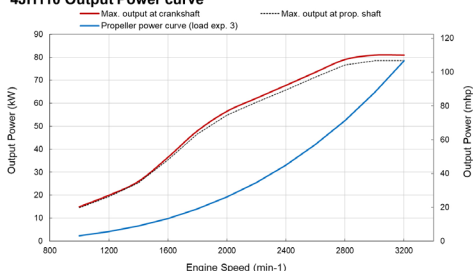
4JH80 Fuel consumption (propload exp 3.0)



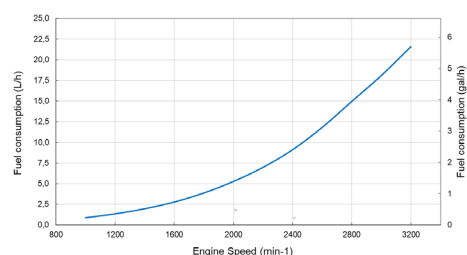
4JH110 Torque curve



4JH110 Output Power curve



4JH110 Fuel consumption (propload exp 3.0)



## MARINE GEAR / DRIVE\*

| Model                                     | KMH4A [down angle: 8°]                         | KM4A2 [down angle: 7°]        | ZF25A [down angle: 8°] | ZF30M [parallel]      | Saildrive [SD60]**                         | Saildrive [SD15]     |
|-------------------------------------------|------------------------------------------------|-------------------------------|------------------------|-----------------------|--------------------------------------------|----------------------|
| Clutch type                               | Hydraulic multi disc                           | Mechanical cone clutch        | Hydraulic multi disc   | Mechanical multi-disc | Mechanical multi-disc                      | Hydraulic multi disc |
| Dry weight                                | 30 kg [66 lbs]                                 | 28.5 kg [63 lbs]              | 33 kg [73 lbs]         | 28 kg [62 lbs]        | 44 kg [97 lbs]                             | 90 kg [198]          |
| Reduction ratio [fwd/asn]                 | 2.04/2.04 2.45/2.45                            | 1.47/1.47 2.14/2.14 2.63/2.63 | 1.93/1.93 2.80/2.80    | 2.15/2.64 2.70/2.64   | 2.23/2.23 2.49/2.49                        | 2.51/2.51            |
| Direction of rotation [viewed from stern] | Clockwise & counterclockwise                   |                               |                        | Clockwise             | Clockwise & counterclockwise               |                      |
| Lubrication system                        | Forced lubrication                             | Splash lubrication            | Forced lubrication     | Splash lubrication    | Splash lubrication                         | Oil bath             |
| Lube oil specifications                   | API class: CD or higher, SAE grade: #20 or #30 |                               | ATF                    |                       | API class: CD or higher, SAE grade: 15W-40 | ATF                  |
| Max. lube oil capacity                    | 2.0 L                                          | 2.0 L                         | 1.8 L                  | 1.1 L                 | 2.8 L                                      | 6 L                  |
| Dry weight - engine & gear                | 259 kg [571 lbs]                               | 257.5 kg [568 lbs]            | 262 kg [578 lbs]       | 257 kg [567 lbs]      | 273 kg [602 lbs]                           | 319 kg [703 lbs]     |

\* Other marine gearboxes and configurations available upon request. Contact your local supplier for more information.

\*\* Not for 4JH110

## ACCESSORIES

## Standard engine package

- Exhaust mixing elbow [L-type]
- Alternator 12V - 125A
- Seawater pump rear mounted
- B25 panel
- Flexible mounts
- Mechanical throttle sensor
- Fuel pre-filter with water separator
- Fresh water expansion tank
- Belt safety cover

## Optional

- High-riser mixing elbow [instead of L-elbow]
- Secondary alternator 12V - 125A or 24V - 60A
- Seawater pump front mounted
- C35 panel
- Extension harness [3m, 5m, 10m]
- Second station
- Sidemount control head
- VC20 Vessel Control System

- NMEA2000 Engine monitoring system
- Semi 2-pole
- SOLAS including Certificate [factory fit only]
- Opposite dipstick
- Hot water kit with cocks
- Extension saildrive leg [75 mm]
- Propeller cone nut kit SD60
- On-board spare parts kit

NOTE: All data subject to change without notice. Text and illustrations are not binding.